

PRACTICAL COURSE WORKBOOK

Ethical Hacking & Penetration Testing

Apply practical protection to accounts and systems

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a prioritised security review and action plan
SKILLS	Ethical hacking, Kali Linux, OWASP, CTF, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Ethical hacking appropriately in a realistic, bounded task.
- Use Kali Linux appropriately in a realistic, bounded task.
- Use OWASP appropriately in a realistic, bounded task.
- Use CTF appropriately in a realistic, bounded task.

WHO THIS IS FOR

Defenders and developers learning lawful, defensive security practice.

BEFORE YOU BEGIN

- Basic networking and operating-system knowledge

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Ethical Hacking & Penetration Testing is a structured, practical course covering Ethical hacking, Kali Linux, OWASP, CTF. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Ethical hacking: Security foundations	1. Assets with Ethical hacking 2. Threats with Kali Linux 3. Risk with OWASP
02 Kali Linux: Defensive controls	1. Identity with CTF 2. Network protection with Ethical hacking 3. Hardening with Kali Linux
03 OWASP: Assessment	1. Safe testing with OWASP 2. Evidence with CTF 3. Remediation with Ethical hacking
04 CTF: Operations	1. Monitoring with Kali Linux 2. Response with OWASP 3. Recovery with CTF

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Ethical hacking: Security foundations

Apply Ethical hacking through assets, threats, risk.

LESSON 01 / 18 MIN

Assets with Ethical hacking

Learning objectives	• Explain assets in the context of Ethical Hacking & Penetration Testing. • Apply Ethical hacking to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	assets, Ethical hacking, security
Retrieval prompt	In Ethical Hacking & Penetration Testing, which evidence best supports a assets result produced with Ethical hacking?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Threats with Kali Linux

Learning objectives	• Explain threats in the context of Ethical Hacking & Penetration Testing. • Apply Kali Linux to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	threats, Kali Linux, security
Retrieval prompt	In Ethical Hacking & Penetration Testing, which evidence best supports a threats result produced with Kali Linux?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Risk with OWASP

Learning objectives	• Explain risk in the context of Ethical Hacking & Penetration Testing. • Apply OWASP to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	risk, OWASP, security
Retrieval prompt	In Ethical Hacking & Penetration Testing, which evidence best supports a risk result produced with OWASP?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Kali Linux: Defensive controls

Apply Kali Linux through identity, network protection, hardening.

LESSON 04 / 30 MIN

Identity with CTF

Learning objectives

- Explain identity in the context of Ethical Hacking & Penetration Testing.
- Apply CTF to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

identity, CTF, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a identity result produced with CTF?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Network protection with Ethical hacking

Learning objectives

- Explain network protection in the context of Ethical Hacking & Penetration Testing.
- Apply Ethical hacking to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

network protection, Ethical hacking, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a network protection result produced with Ethical hacking?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Hardening with Kali Linux

Learning objectives

- Explain hardening in the context of Ethical Hacking & Penetration Testing.
- Apply Kali Linux to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

hardening, Kali Linux, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a hardening result produced with Kali Linux?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

OWASP: Assessment

Apply OWASP through safe testing, evidence, remediation.

LESSON 07 / 26 MIN

Safe testing with OWASP

Learning objectives

- Explain safe testing in the context of Ethical Hacking & Penetration Testing.
- Apply OWASP to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

safe testing, OWASP, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a safe testing result produced with OWASP?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Evidence with CTF

Learning objectives

- Explain evidence in the context of Ethical Hacking & Penetration Testing.
- Apply CTF to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

evidence, CTF, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a evidence result produced with CTF?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Remediation with Ethical hacking

Learning objectives

- Explain remediation in the context of Ethical Hacking & Penetration Testing.
- Apply Ethical hacking to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

remediation, Ethical hacking, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a remediation result produced with Ethical hacking?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

CTF: Operations

Apply CTF through monitoring, response, recovery.

LESSON 10 / 22 MIN

Monitoring with Kali Linux

Learning objectives

- Explain monitoring in the context of Ethical Hacking & Penetration Testing.
- Apply Kali Linux to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

monitoring, Kali Linux, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a monitoring result produced with Kali Linux?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Response with OWASP

Learning objectives

- Explain response in the context of Ethical Hacking & Penetration Testing.
- Apply OWASP to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

response, OWASP, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a response result produced with OWASP?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Recovery with CTF

Learning objectives

- Explain recovery in the context of Ethical Hacking & Penetration Testing.
- Apply CTF to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

recovery, CTF, security

Retrieval prompt

In Ethical Hacking & Penetration Testing, which evidence best supports a recovery result produced with CTF?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Ethical Hacking & Penetration Testing project using Ethical hacking, Kali Linux, OWASP.

DELIVERABLE	A working Ethical Hacking & Penetration Testing artefact plus an evidence-based self-review.
TOOLS	A suitable Ethical hacking environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Ethical hacking.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

Use security techniques only on systems you own or are explicitly authorised to test.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Cybersecurity Framework 2.0

NIST - reviewed 2026-08-21

<https://www.nist.gov/cyberframework>

Web Security Testing Guide

OWASP Foundation - reviewed 2026-08-21

<https://owasp.org/www-project-web-security-testing-guide/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Ethical hacking scope.